

2-Mark Questions & Answers -- MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. List the functional units of the 8086 microprocessor.

The 8086 CPU is divided into two independent functional units:

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Q2. What is memory segmentation in 8086

Memory Segmentation is a technique that divides the 1MB physical memory space into 64KB logical blocks called segments (Code Segment, Data Segment, Stack Segment, Extra Segment). It allows a 16-bit processor to address 20-bit physical addresses using segment base registers and offset registers.

Q3. Define minimum mode and maximum mode of 8086.

- * Minimum Mode (MN/MX' = 1): Single-processor system mode where 8086 generates all bus control signals (RD, WR, ALE, M/IO) directly.

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Q4. What is the purpose of demultiplexing the address/data bus

The 8086 uses multiplexed pins AD0 - AD15 to save physical pin count. Demultiplexing separates the 20-bit memory address from 16-bit data using octal latches (74LS373) driven by the Address Latch Enable (ALE) signal.

Q5. State the size of the 8086 instruction queue.

The 8086 Bus Interface Unit (BIU) contains a 6-byte Instruction Queue that prefetches instruction bytes sequentially from memory when the EU is busy, enabling instruction pipelined prefetching.